

The vertex k -cut problem

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A B S T R A C T

Given an undirected graph $G = (V, E)$, a vertex k -cut of G is a vertex subset of V the removing of which disconnects the graph in at least k components. Given a graph G and an integer $k \geq 2$, the vertex k -cut problem consists in finding a vertex k -cut of G of minimum cardinality. We first prove that the problem is NP-hard for any fixed $k \geq 3$. We then present a compact formulation, and an extended formulation from which we derive a column generation and a branching scheme. Extensive computational results prove the effectiveness of the proposed methods.

Vertex cut
Mixed-integer programming models
Branch and Price
Exact algorithms

1. Introduction

A *vertex cut* of a graph $G = (V, E)$ is a strict subset of vertices $V_0 \subset V$ such that the graph obtained from G by removing V_0 has at least two (non-empty and pair-wise disconnected) components. If the number of components is at least k , the vertex cut V_0 is called a *vertex k -cut*.

Given G and an integer $k \geq 2$, the *vertex k -cut problem* is to find, if it exists, a vertex k -cut of minimum cardinality. Berger, Grigoviev and Zwaan [1] showed that the problem is NP-hard (k being part of the input) but polynomial-time solvable for graphs of bounded treewidth. Ben Ameur and Didi Biha [2] proved that, for $k = 2$, it is polynomial-time solvable as it amounts to computing $|V|^2$ maximum flows. A fixed-parameter algorithm for the vertex k -cut problem, considering the parameter k , would be an algorithm solving the problem with a running time of the form $f(k) \times \text{poly}(|V|)$ where $f(k)$ is any function and $\text{poly}(|V|)$ is a polynomial in $|V|$. Marx [3] showed that such an algorithm is unlikely to exist as he proved the $W[1]$ -hardness of the problem. However, the complexity for fixed k was an open question.

Despite its basic setting, the vertex k -cut problem has received limited attention according to ILP approaches. Let us detail the literature on problem variants, where one wants to minimize the number of vertices to be removed to partition a given graph. The *k-separator problem* is a variant where cardinality bounds are required. It consists in finding a vertex cut whose removal gives a graph where the size of each connected component is less than or equal to k . In [4], the authors analyze the complexity on several classes of graphs. They also propose approximation algorithms, a formulation and a polyhedral study. Another variant of this problem exists where the cardinality constraints are not on the size of the connected components but on vertex sets. More precisely, the problem consists in finding a vertex cut V_0 such that $V \setminus V_0$ can be partitioned into two sets of cardinality less than or equal to k , and no edge is incident to both sets. Remark that each set may contain several connected components. This problem is NP-hard even for planar graphs [5] or maximum degree 3 graphs [6]. A first polyhedral study on this problem is done in [7] from which a Branch-and-Cut algorithm is derived [8]. In [9] the authors introduce valid inequalities based on a lower bound given by the number of disjoint paths between all pairs of vertices. For these inequalities, the authors analyze their facial structures and add these inequalities in a Branch-and-Cut algorithm. In the *q-balanced vertex k-separator problem*, the bound is not on the size of the sets but on their differences. More formally, one seeks for a vertex k -cut V_0 such that $V \setminus V_0$ can be partitioned into k pairwise disconnected sets $V_1, \dots, V_k$ and $|V_i| - |V_j|$ is at most q for all $i \neq j \in \{1, \dots, k\}$. Different integer linear programming formulations are given in [10]. The *multi-terminal vertex k-cut problem* consists, given a set $T \subset V$ of k terminals, in finding a vertex k -cut V_0 of G containing no terminal such that each component of $G[V \setminus V_0]$ contains at most one terminal. In [3] Marx shows also the W[1]-hardness of this problem. A path-based formulation is given in [11] for solving this problem and several inequalities are proposed. A polyhedral analysis is also performed and an efficient Branch-and-Cut algorithm is developed.

Several variants of problems where one wants to disconnect a graph have been considered where an edge set, instead of a vertex set, is removed. Among them, the most two famous problems are the k -cut and the multiway cut problems. The first one is the edge counterpart of our problem: it consists in finding a minimum set of edges the removing of which leads to at least k -connected components [12–14]. In the multiway cut problem [15–20], a vertex subset S is given and one has to find minimum set of edges in order to separate each pair of vertices of S . Some generalizations of this latter problem, such as minimum 0-extension of graph metrics (that also generalizes the minimum (s, t) -cut problem), have been considered. In [21, 22] the classes of graphs for which the minimum 0-extension problem is tractable are presented.

The first contribution of this paper improves on Marx’s results by showing that the vertex k -cut problem is NP-hard for any fixed $k \geq 3$. Our second contribution is to investigate the hardness of the problem in practice, we report computational experiments on the problem using Integer Linear Programming (ILP) tools to solve it on DIMACS instances.

The paper is organized as follows. Section 2 is devoted to the NP-hardness proof of the vertex k -cut problem for any fixed $k \geq 3$. In Section 3, we reformulate this problem as a stable set problem with additional constraints. We deduce a compact integer linear program based on this reformulation. In Section 4, we present a formulation with an exponential number of variables and a polynomial number of constraints. We also give a column generation scheme to solve the linear relaxation. We prove the effectiveness of this approach by showing that the subproblem is polynomial-time solvable by using flow techniques. Section 5 reports the experimental results we obtain by solving the two formulations, the first one by a general-purpose ILP solver and the second by a Branch-and-Price algorithm. The rest of this section is devoted to notation and assumption.

Notation. Throughout, K denotes the set of integers $\{1, \dots, k\}$ and $G = (V, E)$ is a simple undirected graph with $|V| = n$ and $|E| = m$. The complement of a node subset S is denoted $\bar{S} = V \setminus S$, and the complement of G is denoted $\bar{G} = (V, \bar{E})$, where $\bar{E} = \{uv : uv \notin E\}$. We say that u and v are *neighbors* if there is an edge $uv \in E$. A subset $W \subseteq V$ of vertices is a *clique* of G , if any two vertices of W are neighbors,

and it is a *stable set* of G if it is a clique in $\overline{G}$. The cardinality of the largest stable set of G is denoted by $\alpha(G)$. A subset $W \subseteq V$ of vertices is a *vertex k -multiclique* of G , if there is a k -partition $\pi = \{W_1, \dots, W_k\}$ of W such that any two vertices in different sets of π are adjacent in G , with $W_i \neq \emptyset$ for all $i \in \{1, \dots, k\}$. For each $W \subseteq V$, we indicate by $\delta(W)$ the subset of edges incident with exactly one vertex in W (i.e., all edges uv with $u \in W$, $v \in V \setminus W$), and with $E(W)$ the subset of edges incident with two vertices in W (i.e., all edges uv with $u, v \in W$). Finally, we indicate by $\delta(v) \subseteq E$ the subset of edges incident with v .

Assumption. In the rest of the paper, we will assume that $\alpha(G) \geq k$. This is clearly a necessary and sufficient condition for G to have a vertex k -cut. We assume that G is connected. We will also use implicitly the basic property that a vertex k -cut V_0 is a set of vertices such that $V \setminus V_0$ can be partitioned into k non-empty subsets $V_1, \dots, V_k$ that are pairwise disconnected, i.e., there is no edge between two subsets V_i and V_j for all $i \neq j \in \{1, \dots, k\}$.

2. Complexity

In this section, the NP-hardness of the vertex k -cut problem for any fixed $k \geq 3$ is proved.

We start by observing that the problem is equivalent to the *vertex k -multiclique problem* which consists, given an undirected graph $G = (V, E)$ and $k \geq 2$, in determining a vertex k -multiclique of maximum cardinality.

Proposition 1. *A vertex subset V_0 of a graph $G = (V, E)$ is a vertex k -cut if and only if $W = V \setminus V_0$ is a vertex k -multiclique in the complement graph $\overline{G}$.*

We now state our complexity result.

Theorem 1. *For any fixed $k \geq 3$, the vertex k -cut problem is NP-hard.*

Proof. In order to prove the theorem, it suffices to prove that the vertex 3-cut problem is NP-hard. Indeed, G has a vertex k -cut of size s if and only if $\tilde{G}$ has a vertex $(k+1)$ -cut of size s , where $\tilde{G}$ is obtained by adding an isolated vertex to G . The basic idea of the proof is to reduce an instance of the NP-hard maximum stable set problem in tripartite graphs [23] into an instance of the vertex 3-cut problem.

By Proposition 1, it suffices to prove that the vertex 3-multiclique problem is NP-hard. We actually prove that this problem is already NP-hard in the class of tripartite graphs. To this end, we will use a reduction from the maximum stable set problem in tripartite graphs, which is NP-hard by Lemma 6 in [23]. Let $G = (V_1 \cup V_2 \cup V_3, E)$ be a tripartite instance of the maximum stable set problem. Since every isolated vertex belongs to all maximal stable sets, it is still NP-hard to solve tripartite instances with additional isolated vertices, hence, without loss of generality, we can suppose that V_i contains an isolated vertex v_i for each $i \in \{1, 2, 3\}$. We define the instance $\tilde{G} = (V_1 \cup V_2 \cup V_3, \tilde{E})$ of the 3-multiclique problem where $\tilde{E} = \{uv : u \in V_i, v \in V_j, i \neq j, uv \notin E\}$. (In Fig. 1, the white vertices represent a maximal stable set of G (left graph). The same set corresponds to a maximal 3-multiclique on $\tilde{G}$ (right graph).)

We claim that a vertex subset S is a stable set of G containing $\{v_1, v_2, v_3\}$ if and only if S is a vertex 3-multiclique of $\tilde{G}$ containing $\{v_1, v_2, v_3\}$. Indeed, by construction, two vertices $u \in V_i \cap S$ and $v \in V_j \cap S$ where $i \neq j$ are adjacent in $\tilde{G}$. Thus S is a vertex 3-multiclique in $\tilde{G}$. The converse is also true. Since any maximum stable set of G and any maximum vertex 3-multiclique of $\tilde{G}$ contain $\{v_1, v_2, v_3\}$, the proof is done. $\square$

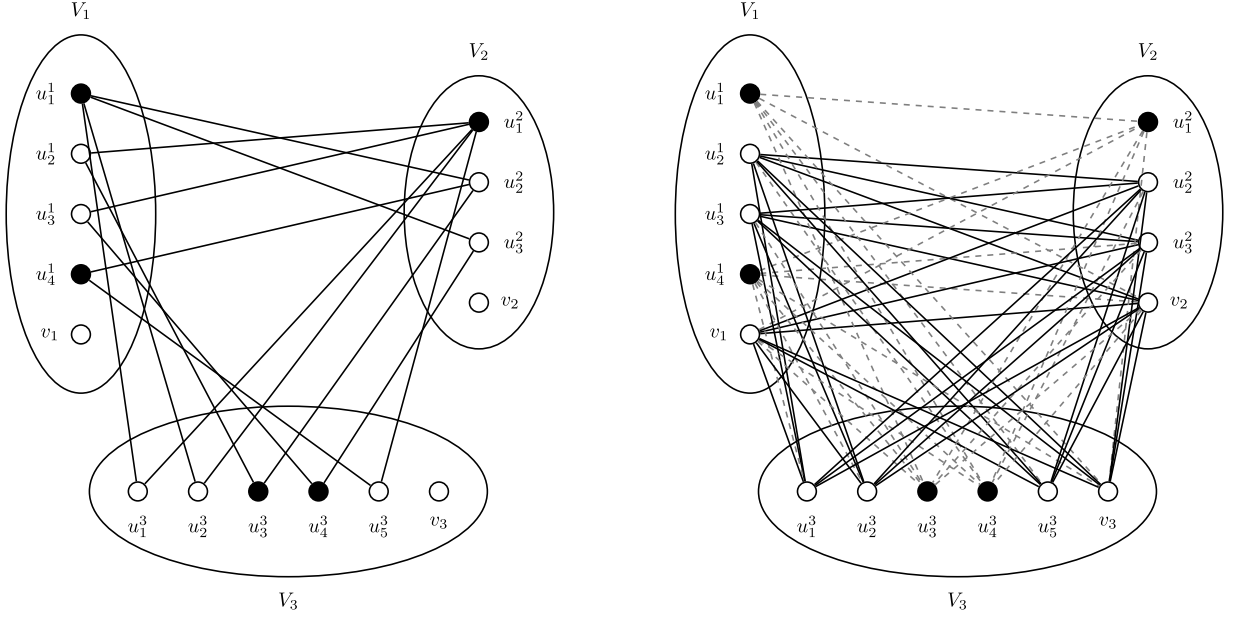


Fig. 1. Reduction from the maximum stable set problem in tripartite graphs to the 3-multiclique problem.

3. Compact formulation

In this section, we show that the vertex k -cut problem can be reformulated as a maximum stable set problem on a specific k -partite graph with additional requirements. We also derive a compact integer linear program based on this reformulation.

Let $G = (V, E)$ and $k \geq 2$ be an instance of the vertex k -cut problem. As previously noted, a subset $V_0 \subset V$ is a vertex k -cut of G if and only if $V \setminus V_0$ can be partitioned into k nonempty pairwise disconnected sets. Hence, the vertex k -cut problem is equivalent to finding k nonempty disjoint sets $V_1, \dots, V_k$ of V which are pairwise disconnected such that $|\bigcup_{i \in K} V_i|$ is maximum.

We construct a k -partite graph $G' = (V', E')$ so that the vertex k -cut problem on G reduces to the maximum stable set on G' . [Fig. 2](#) gives an illustration of this equivalence. The graph on the left is G . The set V_0 of white vertices corresponds to a 3-vertex cut and $\{V_1, V_2, V_3\}$ with $V_1 = \{v_3\}$, $V_2 = \{v_4\}$ and $V_3 = \{v_2, v_5\}$ is a partition of $V \setminus V_0$ into 3 pairwise disconnected sets. The graph on the right corresponds to G' . The white vertices form the stable set associated with $\{V_1, V_2, V_3\}$.

Formally the construction of G' is as follows. The set V' is obtained by considering k copies $v^1, \dots, v^k$ of every vertex $v \in V$. We define the k -partition of V' as $\pi = \{V'_1, \dots, V'_k\}$ with $V'_i = \{v^i : v \in V\}$ for all $i = 1, \dots, k$. In other words, V'_i corresponds to a copy of V . The edge set E' is the union of two sets E'_α and E'_β . $E'_\alpha = \{v^i v^j : i \neq j \in K\}$ is the edge set obtained by considering a clique between all the copies of a same vertex $v \in V$. For E'_β , we consider for each $uv \in E$ an edge between every copy of u and every copy of v , except for the copies belonging to the same cluster of π . Hence, $E'_\beta = \{u^i v^j : uv \in E, i \neq j \in K\}$. There is a 1-to-1 correspondence between nonempty pairwise disconnected disjoint sets $V_1, \dots, V_k$ of V and stable sets of G' intersecting each V'_i , $i \in K$. Indeed, let $V_1, \dots, V_k$ satisfying the aforementioned requirements. Let $S \subseteq V'$ be the set obtained by taking in V'_i the copies of the vertices in V_i for all $i \in K$. S is a stable set because no edge exists between V_i and V_j and $V_i \cap V_j = \emptyset$ for $i \neq j \in K$. Moreover, S intersects every V'_i , $i \in K$, since $V_1, \dots, V_k$ are nonempty. Finally $S = |\bigcup_{i \in K} V_i|$. The converse also holds which implies the result.

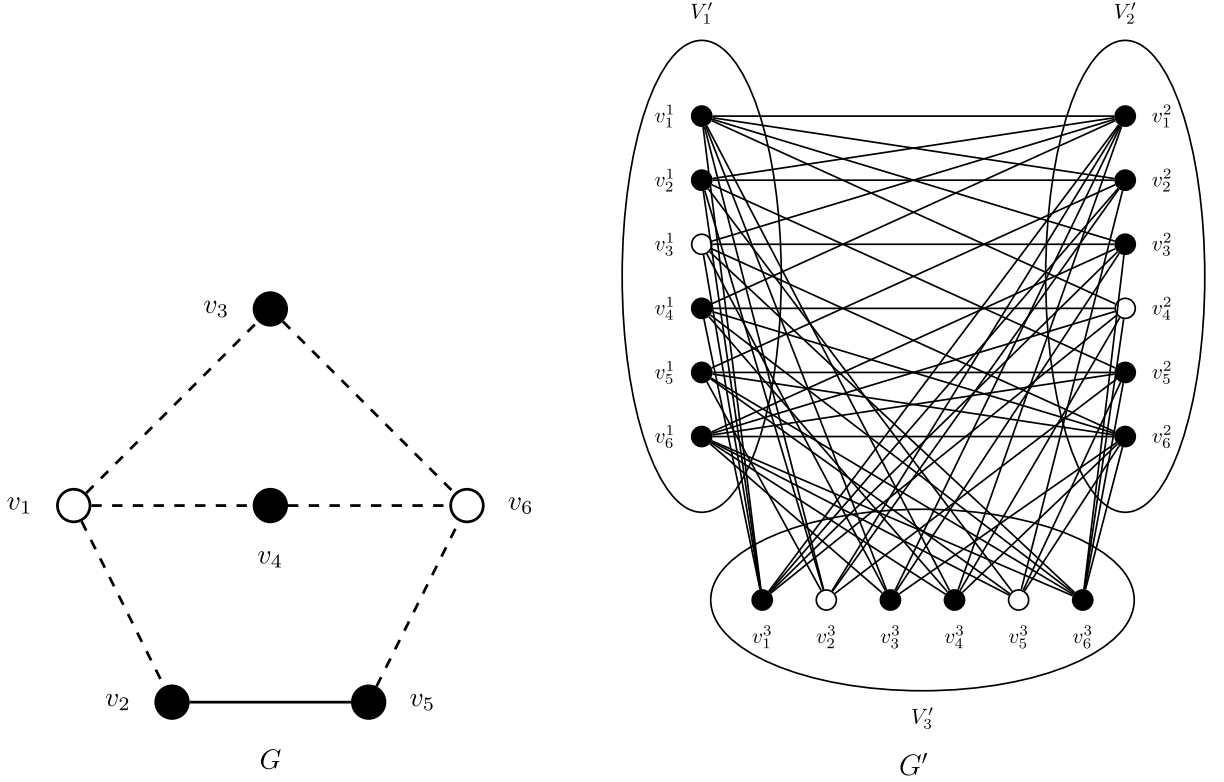


Fig. 2. Transformation of the 3-vertex cut problem into a specific maximum stable set problem.

We now give a formulation of the vertex k -cut problem with an integer linear program. By the previous reformulation, we look for a stable set S of G' intersecting every V'_i of the k -partition. For all vertices $v \in V$ and for all integer $i \in K$, let us associate a binary variable x_v^i such that:

$$x_v^i = \begin{cases} 1 & \text{if copy } v^i \in V'_i \text{ of vertex } v \in V \text{ belongs to } S \\ 0 & \text{otherwise} \end{cases} \quad i \in K, v \in V.$$

The first natural compact ILP formulation (called ILP_C) reads as follows:

$$(\text{ILP}_C) \quad \max \sum_{i \in K} \sum_{v \in V} x_v^i \quad (1)$$

$$\sum_{i \in K} x_v^i \leq 1 \quad v \in V, \quad (2)$$

$$x_u^i + \sum_{j \in K \setminus \{i\}} x_v^j \leq 1 \quad i \in K, uv \in E, \quad (3)$$

$$\sum_{v \in V} x_v^i \geq 1 \quad i \in K, \quad (4)$$

$$x_v^i \in \{0, 1\} \quad i \in K, v \in V. \quad (5)$$

The objective function maximizes the size of S . Constraints (2) and (3) are the clique constraints associated with cliques of E'_α and edges of E'_β , respectively. Constraints (4) impose that S intersects each V'_i for $i \in K$.

By replacing constraints (5) with

$$x_v^i \geq 0 \quad i \in K, v \in V, \quad (6)$$

we obtain the Linear Programming relaxation of ILP_C , that will be denoted as LP_C in what follows. Descriptive natural ILP models are known to produce weak linear programming relaxations as the following proposition shows:

Proposition 2. *An optimal solution to LP_C is $x_v^i = \frac{1}{k}$, $i = 1, \dots, k$, $v \in V$, and has value $n = |V|$.*

Proof. Constraints (2) impose a trivial upper bound of value n . By setting $x_v^i = \frac{1}{k}$, $i = 1, \dots, k$, $v \in V$, the objective function obtains exactly the value n and all the other constraints are satisfied by construction. $\square$

In order to improve the strength of the linear programming relaxation, and to remove the symmetry of ILP_C , in the next section we design a new formulation for the vertex k -cut problem.

4. Exponential-size formulation

In this section, we derive an alternative formulation for the vertex k -cut problem having an exponential number of variables with respect to the input size. Let $\mathcal{S} = \{S \subseteq V, S \neq \emptyset\}$ be the family of all non-empty subsets of vertices of V .

For all subsets $S \in \mathcal{S}$, let us associate a binary variable ξ_S such that:

$$\xi_S = \begin{cases} 1 & \text{if } S \text{ corresponds to one of the } k \text{ pair-wise disconnected subsets of } G \\ 0 & \text{otherwise} \end{cases} \quad S \in \mathcal{S}.$$

The vertices that do not appear in any selected subset are assigned to the vertex cut. In the following we let $\mathcal{C}$ be an edge-covering family of cliques of G , that is, a family of cliques so that for each edge $uv \in E$, there is at least one clique $C \in \mathcal{C}$ containing both $u, v \in C$. The exponential-size ILP formulation for the vertex k -cut problem reads as follows

$$(\text{ILP}_E) \quad \max \sum_{S \in \mathcal{S}} |S| \xi_S \quad (7)$$

$$\sum_{S \in \mathcal{S}: v \in S} \xi_S \leq 1 \quad v \in V, \quad (8)$$

$$\sum_{S \in \mathcal{S}: C \cap S \neq \emptyset} \xi_S \leq 1 \quad C \in \mathcal{C}, \quad (9)$$

$$\sum_{S \in \mathcal{S}} \xi_S = k \quad (10)$$

$$\xi_S \in \{0, 1\} \quad S \in \mathcal{S}. \quad (11)$$

The objective function (7) maximizes the sum of the cardinalities of the selected subsets S of vertices, which is equivalent to minimize the cardinality of the vertex cut. Constraints (8) impose that each vertex $i \in V$ does not appear in more than one of the selected subsets. Constraints (9) impose that, for each clique $C \in \mathcal{C}$, at most one subset containing any vertex of the clique can be selected. Constraint (10) imposes that exactly k subsets are selected. Constraints (11) impose the variables to be binary, so, finally, by relaxing the integrality of constraints (11) to

$$\xi_S \geq 0 \quad S \in \mathcal{S}, \quad (12)$$

we obtain the Linear Programming relaxation of ILP_E , that is denoted as LP_E in what follows.

4.1. A Branch-and-Price algorithm

In this section we describe a Branch-and-Price algorithm which is designed to solve ILP_E . The exact algorithm is composed by two main components, i.e., a Column Generation (CG) algorithm to solve LP_E , and a branching scheme. We treat these two aspects in the next sections.

4.1.1. Solving the linear programming relaxation of ILP_E

Model (7)–(11) has exponential size, thus a *column generation* procedure is necessary to solve LP_E .

The *master problem* (MP) can be initialized with the n subsets of V containing a single vertex. Since we assumed that G contains a stable set of cardinality k , this initialization assures the existence of a feasible solution to start the column generation. Additional variables needed to optimally solve the MP are then generated by separating the associated dual constraints. The *pricing problem* (PP) (see, e.g., [24] for definition and more details on column generation) can be solved efficiently as described in the following.

At each column generation step, the optimal values $\lambda^* \in \mathbb{R}_+^V$, $\pi^* \in \mathbb{R}_+^{\mathcal{C}}$, $\gamma^* \in \mathbb{R}$ (respectively) of the dual variables associated with constraints (8), (9), (10) (respectively) are given. The separation of a violated dual constraint is equivalent to find a non-empty subset $S^* \in \mathcal{S}$ such that

$$\sum_{v \in S^*} \lambda_v^* + \sum_{C \in \mathcal{C}: C \cap S^* \neq \emptyset} \pi_C^* + \gamma^* < |S^*|$$

which can be reformulated as

$$\sum_{v \in S^*} \nu_v^* - \sum_{C \in \mathcal{C}: C \cap S^* \neq \emptyset} \pi_C^* > \gamma^*, \quad (13)$$

where $\nu_v^* = 1 - \lambda_v^*$.

If such a subset exists, the corresponding variable ξ_{S^*} is added to the MP, and the procedure is iterated; otherwise, the MP is solved to proven optimality. Hence PP amounts to find a S^* maximizing the left-term in (13) and to check whether or not it is bigger or not than the right-term. It can be modeled as a Binary Linear Program using variables x_v ($v \in V$), which define S^* , and variables y_C ($C \in \mathcal{C}$), each of which takes value 1 if clique C intersects set S^* , as follows:

$$\max \sum_{v \in V} \nu_v^* x_v - \sum_{C \in \mathcal{C}} \pi_C^* y_C \quad (14)$$

$$y_C \geq x_v \quad v \in C \in \mathcal{C}, \quad (15)$$

$$\sum_{v \in V} x_v \geq 1 \quad (16)$$

$$x_v \in \{0, 1\} \quad v \in V, \quad (17)$$

$$y_C \in \{0, 1\} \quad C \in \mathcal{C}. \quad (18)$$

Constraints (15) impose $y_C = 1$ ($C \in \mathcal{C}$) if at least a vertex v of a clique C belongs to S^* ; while constraints (16) impose S^* is not empty. If the value of the optimal solution of the PP is larger than γ^* , $S^* = \{v \in V, x_v^* = 1\}$, and the associated variable z_{S^*} is added to the MP. Note that, since $\pi_C \geq 0$ ($C \in \mathcal{C}$) and variables x_v ($v \in V$) are binary, we can relax constraints (18) to $y_C \geq 0$ ($C \in \mathcal{C}$).

The PP can be interpreted as follows: Given $G = (V, E)$, a profit ν_v^* for each $v \in V$ (possibly negative) and a penalty $\pi_C^* \geq 0$ for each $C \in \mathcal{C}$, the problem aims at selecting a non-empty subset of vertices of maximum profit; the penalty π_C^* associated with a clique C is paid if at least one of its vertices is selected. A vertex v with $\nu_v^* \leq 0$ can be removed together with its incident edges. If a clique $C \in \mathcal{C}$ is reduced to a single vertex u by the removal, then, π_C is subtracted from the profit ν_u^* of vertex u . The procedure is

iterated until all vertices have positive profit. In case all vertices have negative profit ν^* , or all vertices are removed, the PP problem reduces to finding the vertex $u = \arg \max_{v \in V} \{\nu_v^* - \sum_{C \in \mathcal{C}: v \in C} \pi_C^*\}$.

The following proposition characterizes the complexity of the PP.

Proposition 3. *The PP is polynomial-time solvable.*

Proof. In the PP we are looking for a non-empty subset of vertices. Let us define $\text{PP} \cup \emptyset$ a relaxation of the PP, where also the empty set is admitted as solution. Given a polynomial-time algorithm for the $\text{PP} \cup \emptyset$, we can select a vertex $v \in V$ which is forced to be in S^* , and then apply the algorithm to the subgraph of G induced by $V \setminus \{v\}$. By applying this procedure for each $v \in V$, in n iterations we obtain the optimal solution to the PP.

It remains to show that $\text{PP} \cup \emptyset$ is polynomially solvable. This can be done by observing that the $\text{PP} \cup \emptyset$ can be formulated by removing constraint (16) from model (14)–(18). The resulting model structure is the same as the generic structure described in [25], model (8). A solution method for the latter model is given in [25]. (The latter consists in solving a min-cut/max-flow problem on a network with one node associated with each variable, plus a source and a sink node as explained below.) $\square$

In the case of our problem, we have a network $N = (W, A)$, where the node set is $W = \{s, t\} \cup V \cup \{u_C, C \in \mathcal{C}\}$, i.e., in addition to the vertex set V of G , there are a source and a sink node, and a node u_C for each clique $C \in \mathcal{C}$.

The arc set is defined below:

- For each $v \in V$ there is an arc (s, v) with capacity ν_v^* ,
- For each $C \in \mathcal{C}$ there is an arc (C, t) with capacity π_C^* ,
- For each $C \in \mathcal{C}$ and each $v \in C$, there is an arc (v, C) with infinite capacity.

There is a one-to-one correspondence between st -cuts of finite capacity in N , and feasible solutions of (14)–(18). For a cut of capacity L , the corresponding solution of (14)–(18) has value $\sum_{C \in \mathcal{C}} \pi_C^* - L$. Hence, an optimal solution to (14)–(18) can be obtained from a min-cut on a network N having $n + |\mathcal{C}| + 2$ nodes.

Corollary 1. *An optimal solution to the linear relaxation LP_E of the exponential-size integer formulation ILP_E of the minimum vertex k -cut problem can be computed in polynomial time.*

Proof. Since the pricing problems PP ask for the solution of $n = |V|$ min-cut problems, then solving the master MP (and, eventually, LP_E) is polynomial time solvable. $\square$

In case $\mathcal{C}$ is exactly the set of edges of the graph G (which is the only possible form of $\mathcal{C}$ for a triangle free graph), a solution method based on solving a min-cut problem on a smaller network having $n + 2$ nodes is described in the Appendix. Finally, let us mention that the constraint matrix of (15) is totally unimodular, as observed in [10] for the case of edge constraints. This gives an alternative proof of polynomial-time solvability for the PP.

4.1.2. A branching scheme for ILP_E

When the optimal solution of the master problem (MP) associated with the linear relaxation of model ILP_E for the min vertex k -cut problem is fractional, a branching scheme is necessary in order to obtain an integer solution.

Let ξ^* be the current (fractional) solution of the MP. A two-level branching scheme has to be considered. First we branch by imposing that, for each vertex $v \in V$, either v is in the vertex k -cut V_0 or it belongs to

the vertex-set S of some component of the subgraph of G induced by $V \setminus V_0$. This is in general not enough to define an integer solution, indeed, even if the vertex k -cut V_0 is well defined by

$$V_0 = \left\{ v \in V : \sum_{S \in \mathcal{S}: v \in S} \xi_S^* = 0 \right\},$$

it does not imply that the solution is 0–1 valued. We impose a second level branching, where, for two vertices u and v outside V_0 , we impose that either u and v are in the same component, or they belong to different ones.

In the first branching, for each vertex $v \in V$, we check if it is partially included in the components and the vertex cut, more precisely, if

$$0 < \sum_{S \in \mathcal{S}: v \in S} \xi_S^* < 1. \quad (19)$$

In case of multiple partially included vertices, we branch on the vertex v for which the sum in (19) is closer to 1. Ties are broken randomly. Two subproblems are then created from the current one:

- in the first subproblem, we impose that v is in the vertex cut, by modifying the associated constraint (8) to

$$\sum_{S \in \mathcal{S}: v \in S} \xi_S = 0;$$

we also modify the pricing procedure in order to forbid the selection of vertex v by modifying the cost structure of the associated min-cut problem;

- in the second subproblem, we impose that v is not in the vertex cut, by modifying the associated constraint (8) to

$$\sum_{S \in \mathcal{S}: v \in S} \xi_S = 1;$$

the pricing procedure is unchanged.

Once V_0 is defined, then ξ^* is still fractional if and only if we can find two vertices u, v so that

$$0 < \sum_{S \in \mathcal{S}: u, v \in S} \xi_S^* < 1. \quad (20)$$

(It holds more generally for 0–1 constraints of the form $A\xi^* = \mathbf{1}$, see [26].)

In case more than one such pair of vertices exist, we branch on the pair for which the sum in inequality (20) is closer to 1. Ties are broken randomly. Two subproblems are then created from the current one:

- in the first subproblem, we impose that u and v are in the same component; this can be obtained by contracting $\{u, v\}$ in the pricing subproblem, that is, creating a supervertex w representing both u and v and such that $\delta(w) = \delta(u) \cup \delta(v)$ (and removing u, v);
- in the second subproblem, we impose that u and v are in different components; this can be obtained by adding to the pricing subproblem an incompatibility constraint between u and v . In this case the subproblem cannot be formulated as a min-cut/max-flow problem, and we have to solve the MIP formulation (14)–(18) with the additional constraint

$$x_v + x_u \leq 1.$$

In this case the modified pricing problem might be NP-hard.

In our Branch-and-Price algorithm we first define the vertices in the vertex cut, i.e., we apply the first branching rule. Then, in case the solution is still fractional, we apply the second branching rule. After branching, the variables that are incompatible with the branching decision are removed from the children nodes. The following proposition states that the two proposed branching rules define a complete branching scheme for ILP_E :

Proposition 4. *The two branching rules applied in sequence provide a complete branching scheme for model ILP_E .*

Proof. The rows of the constraints (8) associated with vertices forced out of the vertex cut, after the application of the first branching rule, are equalities with binary coefficients and right-hand-side equal to 1. In this case, if a basic solution ξ^* is fractional, then there exist u and v such that (20) holds. This result allows to conclude that, if a solution is fractional after the first branching rule is applied, then we can determine two vertices for applying the second branching rule. $\square$

4.2. Comparison of the strength of the LP relaxations of the two formulations

This section discusses the relation between the two formulations proposed for the vertex k -cut problem.

Proposition 5. *Even when $\mathcal{C} = E$, the bound for the vertex k -cut problem provided by the optimal solution value of the extended formulation LP_E strictly dominates the corresponding bound provided by the compact formulation LP_C .*

Proof. Given a feasible solution $\tilde{\xi}$ of LP_E , we can construct a feasible solution $\tilde{x}$ of LP_C with same objective function value as follows:

$$\tilde{x}_v^i = \frac{1}{k} \sum_{S \in \mathcal{S}: v \in S} \tilde{\xi}_S \quad i \in K, v \in V.$$

We first show that the two solutions have the same objective function value:

$$\sum_{i \in K} \sum_{v \in V} \tilde{x}_v^i = \sum_{k \in K} \frac{1}{k} \sum_{v \in V} \sum_{S \in \mathcal{S}: v \in S} \tilde{\xi}_S = \sum_{S \in \mathcal{S}} \sum_{v \in S} \tilde{\xi}_S = \sum_{S \in \mathcal{S}} |S| \tilde{\xi}_S.$$

It is straightforward to check that constraints (2) are satisfied. For each edge $uv \in E$ and for $i \neq j \in K$:

$$\tilde{x}_u^i + \sum_{j \in K \setminus \{i\}} \tilde{x}_v^j = \left(\frac{1}{k} \sum_{S \in \mathcal{S}: u \in S} \tilde{\xi}_S + \frac{k-1}{k} \sum_{S \in \mathcal{S}: v \in S} \tilde{\xi}_S \right) \leq 1,$$

i.e., constraints (3) are satisfied. Finally constraints (4) are satisfied since for each $i \in K$:

$$\sum_{v \in V} \tilde{x}_v^i = \sum_{v \in V} \frac{1}{k} \sum_{S \in \mathcal{S}: v \in S} \tilde{\xi}_S \geq \frac{1}{k} \sum_{S \in \mathcal{S}} \tilde{\xi}_S = 1$$

To see that the domination can be strict, consider now solving the vertex k -cut problem with $k = 3$ for a cycle of 6 vertices. An optimal solution to LP_C is $x_v^i = \frac{1}{3}$, $v \in V$, $i = 1, \dots, 3$, with value 6, while an optimal solution to LP_E has value 3. $\square$

In the remaining of this section we discuss with an example the quality of the linear relaxation of ILP_E , when constraints (9) are expressed for a family of cliques $\mathcal{C}$ or for the edge set E , respectively.

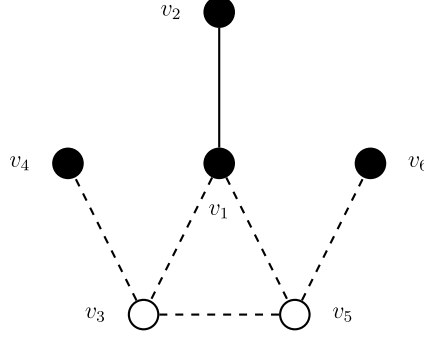


Fig. 3. A graph G and an optimal solution to the vertex k -cut problem with $k = 3$. Removing the white vertices disconnects G in 3 components.

Let us consider the graph $G = (V, E)$ of Fig. 3. The example graph has 6 vertices $(v_1, v_2, v_3, v_4, v_5, v_6)$ and 6 edges $(v_1v_2, v_1v_3, v_1v_5, v_3v_4, v_4v_5, v_5v_6)$. One optimal solution to the min vertex 3-cut problem is obtained by removing vertices v_3 and v_5 , and the maximum in ILP_E is then 4. By defining the clique family $\mathcal{C} = \{\{v_1, v_3, v_5\}, \{v_1, v_2\}, \{v_3, v_4\}, \{v_5, v_6\}\}$, the optimal solution of LP_E is integer of value 4 and it is given by $\xi_{S_1} = \xi_{S_2} = \xi_{S_3} = 1$ where $S_1 = \{v_4\}, S_2 = \{v_6\}, S_3 = \{v_1, v_2\}$. If we consider instead $\mathcal{C} = E$, the optimal solution of LP_E is not integer and has value 4.5. This second solution is given by $\xi_{S_1} = \xi_{S_2} = \xi_{S_3} = \xi_{S_4} = \xi_{S_5} = \xi_{S_6} = 0.5$ where $S_4 = \{v_2\}, S_5 = \{v_5, v_6\}, S_6 = \{v_3, v_4\}$. This example shows a case where a strictly better bound is obtained by considering maximal cliques in $\mathcal{C}$.

5. Computational experiments

Despite the relevance of the vertex k -cut problem, to the best of our knowledge, no previous computational study on exact approaches appeared in the literature. Thus, lacking a previous approach to compare the performance of our Branch-and-Price, with these experiments we wish to assess what is the size of instances that can be tackled by a “standard” approach, relying on the use of a state-of-the-art MIP solver, and whether an approach based on an extended formulation can improve on the standard one, or can eventually be complementary to the standard one, that is, by effectively solving instances with different features. Indeed, the development of a tailored solution approach (as our Branch-and-Price algorithm) is justified when it can enlarge the set of instances for which an optimal solution can be computed, or at least can improve information on upper or lower bounds. With this in mind, we first tune the performance of the two approaches on two sets of widely used graph instances from the literature, and then consider a further set of instances specifically collected for variants of vertex separator problems. In detail, in our experiments we evaluate:

- The computational performance of the compact formulation ILP_C of Section 3, solved via a general purpose ILP solver;
- The computational performance of the extended formulation ILP_E of Section 4, solved via the Branch-and-Price algorithm described in Section 4.1;
- The size of solvable vertex k -cut problem instances, in terms of number of vertices of the graph;
- The effect of the number of subsets k of the partition on the relative performance of the two mentioned exact methods.

Experimental setting. The compact formulation ILP_C is enhanced by a preprocessing phase in which a subset of variables is removed so as to reduce the symmetry of the formulation and to improve the quality

of the associated linear programming relaxation. In this preprocessing, we search for $k - 1$ vertex-disjoint cliques $C_1, \dots, C_i, \dots, C_{k-1}$ of the graph G , and remove the following variables

$$x_v^h, \quad i = 1, \dots, k - 1, \quad v \in C_i, \quad h = i + 1, \dots, k. \quad (21)$$

Indeed, two vertices u, v of a clique cannot be in two different subsets V_i and V_j . Then for all solutions we can reorder the sets $V_1, \dots, V_k$ to ensure that each vertex of a clique C_i must be in one set V_j $j \leq i$ or in the separator. Thus we can remove the variables (21) to reduce the symmetry. The resulting model is then solved by using the MIP solver of **Cplex** 12.6.0 in single-thread mode and default parameter setting. The resulting solution method is denoted as **Cplex + reduction** in what follows.

The extended formulation ILP_E is solved via the Branch-and-Price algorithm, initialized with n variables ξ_S , where $S = \{v\}$, $v \in V$. At each column-generation iteration, linear programs are solved with **Cplex** 12.6.0. The pricing subproblem, formulated as a min-cut/max-flow problem, is solved by means of the pre-flow algorithm by Goldberg and Tarjan [27]. We very rarely observed a branching requiring to solve the subproblem as a MIP (i.e., introducing incompatibility constraints between vertices). The exploration of the branching tree is performed in a depth-first fashion.

The experiments have been performed on a computer with a 3.40 Ghz 8-core Intel Core i7-3770 processor and 16Gb RAM, running a 64 bits Linux operating system. Both exact approaches were tested with a time limit of 3600 s of computing time.

Test-bed of instances. In the computational experiments, we considered one class of classical graph instances from DIMACS challenges, divided into two sets, and an additional class of instances specifically collected for variants of vertex separator problems, divided into three sets. All considered instances having up to 150 vertices. For the first class of instances, that are also used for tuning the solution approaches, we only consider graphs for which the size of the largest stable $\alpha(G)$ is at least 5. For the latter class, we only consider instances for which $\alpha(G)$ is at least 15. Instances are listed in Table 1. In the table, after the instance name, we report the number of vertices n , the number of edges m , the density d , and the size of largest stable set in the graph $\alpha(G)$. This last parameter determines whether a graph instance is feasible for a given value of k , i.e., $\alpha(G) \geq k$; and the corresponding stable set provides a feasible vertex k -cut problem solution.

The first class is composed by a set of 42 instances originally proposed for Maximum Clique, Graph Coloring, and Satisfiability in the second DIMACS challenge [28]. They have from 11 to 149 vertices, with densities varying from 3.35 to 96.79. The $\alpha(G)$ parameter varies from 5 to 80. It includes a second set, composed by 7 instances originally proposed for Graph Partitioning and Graph Clustering in the tenth DIMACS challenge [28]. They have from 34 to 115 vertices, with densities varying from 6.84 to 22.94. The $\alpha(G)$ parameter varies from 17 to 53.

The second class is composed by three sets of instance proposed in [8] and collects intersection graphs of the coefficient matrices of systems of linear equations from different applications, including Physics, Electrical Engineering, Meteorology, Economics and Mathematics. The intersection graph of a matrix has one vertex for each column of and an edge between a pair of vertices if there exists an equation in the system where both variables have a nonzero coefficient. When the linear system is solved by some decomposition method, it will be divided into smaller subsystems that are solved separately. However, the solution of the whole system asks for merging the solutions of the subsystems (i.e., the same variables must take the same values in all the subsystems), and the cost of this tasks increases with the number of variables that appear in more than one subsystem. If one wants to partition the equations into k subsystems, the problem of minimizing the number of common variables can be formulated as a vertex k -cut problem. The instances in the second class, denoted as Intersection Graph Instances in the following, have from 49 to 136 vertices, with densities varying from 5.03 to 59.72. The $\alpha(G)$ parameter varies from 15 to 45. For more details on the second class, the reader is referred to [8].

Table 1
Instance features.

	n	m	d	$\alpha(G)$		n	m	d	$\alpha(G)$
2nd-DIMACS					2nd-DIMACS				
myciel3	11	20	36.36	5	myciel6	95	755	16.91	47
myciel4	23	71	28.06	11	queen8_12	96	1368	30.00	8
queen5_5	25	160	53.33	5	mug100_1	100	166	3.35	33
1-FullIns_3	30	100	22.99	14	mug100_25	100	166	3.35	33
queen6_6	36	290	46.03	6	queen10_10	100	1470	29.70	10
2-Insertions_3	37	72	10.81	18	4-FullIns_3	114	541	8.40	55
myciel5	47	236	21.83	23	games120	120	638	8.94	22
queen7_7	49	476	40.48	7	queen11_11	121	1980	27.27	11
2-FullIns_3	52	201	15.16	25	r125.1	125	209	2.70	49
3-Insertions_3	56	110	7.14	27	DSJC125.1	125	736	9.50	34
queen8_8	64	728	36.11	8	r125.5	125	3838	49.52	5
1-Insertions_4	67	232	10.49	32	DSJC125.5	125	3891	50.21	10
huck	74	301	11.14	27	r125.1c	125	7501	96.79	7
4-Insertions_3	79	156	5.06	39	miles250	128	387	4.76	44
jean	80	254	8.04	38	miles500	128	1170	14.39	18
3-FullIns_3	80	346	10.95	37	miles750	128	2113	26.00	12
queen9_9	81	1056	32.59	9	miles1000	128	3216	39.57	8
david	87	406	10.85	36	miles1500	128	5198	63.95	5
mug88_1	88	146	3.81	29	anna	138	493	5.22	80
mug88_25	88	146	3.81	29	queen12_12	144	2596	25.21	12
1-FullIns_4	93	593	13.86	45	2-Insertions_4	149	541	4.91	74
10th-DIMACS					10th-DIMACS				
karate	34	78	13.90	20	polbooks	105	441	8.08	43
chesapeake	39	170	22.94	17	adjnoun	112	425	6.84	53
dolphins	62	159	8.41	28	football	115	613	9.35	21
lesmis	77	254	8.68	35					
MM-I					MM-HD				
bcspwr02	49	177	15.05	16	L80.cavity01	80	1201	38.01	31
impcol	59	329	19.23	20	L80.wm1	80	1786	56.52	15
dwt	59	256	14.96	15	L100.cavity01	100	1844	37.25	36
can62	62	210	11.11	18	L100.wm1	100	2956	59.72	17
dwt72	72	170	6.65	24	L100.wm3	100	2934	59.27	15
ash219	85	219	6.13	29	L120.cavity01	120	2972	41.62	36
dwt87	87	726	19.41	16	L120.wm2	120	3387	47.44	23
MM-II					MM-II				
ash331	104	331	6.18	30	L125.gre	125	1177	15.19	19
gre	115	576	8.79	33	L125.lop163	125	1218	15.72	17
bcspwr03	118	576	8.34	32	L125.dwt	125	943	12.17	16
L125.will199	125	386	4.98	45	L125.can	125	1257	16.22	15
L125.west0167	125	444	5.73	39	west0132	132	560	6.48	39
L125.ash608	125	390	5.03	37	rw136	136	641	6.98	39
L125.can	125	1022	13.19	20					

Computational performance on the DIMACS instances. In [Tables 2](#) and [3](#) we consider values of $k = 5, 10, 15, 20$, and report, for **Branch and Price** and **Cplex + reduction**, the CPU time is seconds (tl for time limit) and the associated number of explored nodes. For each instance and for each value of k , we report in bold the fastest method. Missing lines correspond to infeasible instances. At the end of each block, we report the number of instances solved to optimality by each method, with respect to the total.

- For $k = 5$, there are 42 2nd-DIMACS and 7 10th-DIMACS instances, 49 feasible instances in total. For 9 instances, no method could find the optimal solution within time limit; the **Branch and Price** could solve 26 out of 49 instances and is the fastest method in 9 cases; the **Cplex + reduction** could solve 40 out of 49 instances and is the fastest method in 30 cases; 13 instances are solved by **Cplex + reduction** while **Branch and Price** fails. For the solved instances, the number of nodes explored by the **Branch and Price** is not larger than 9651 but typically smaller than 100, **Cplex + reduction** in

Table 2Formulation performance comparison on the DIMACS instances ($k = 5$ and $k = 10$).

	$k = 5$				$k = 10$			
	Branch and Price		Cplex + reduction		Branch and Price		Cplex + reduction	
	Time	Nodes	Time	Nodes	Time	Nodes	Time	Nodes
myciel3	0.00	5	0.00	41				
myciel4	0.15	24	0.30	330	0.05	26	1.88	2 241
queen5_5	0.05	34	0.01	0				
1-FullIns_3	0.99	45	0.36	229	0.21	24	1.76	2 464
queen6_6	0.92	304	1.18	577				
2-Insertions_3	0.35	26	1.13	1 210	0.03	2	60.79	43 344
myciel5	782.73	38	2.50	1 423	95.34	106	48.98	12 739
queen7_7	1941.08	9651	40.04	17 950				
2-FullIns_3	<i>tl</i>	543	2.52	1 481	66.57	117	235.00	160 892
3-Insertions_3	6.59	38	9.78	5 848	0.49	15	<i>tl</i>	1 343 150
queen8_8	<i>tl</i>	6729	904.87	365 825				
1-Insertions_4	1659.91	82	6.30	2 139	12.74	29	1242.08	394 816
huck	0.20	6	0.03	0	0.16	8	5.90	4 745
4-Insertions_3	76.38	54	24.41	11 224	6.11	32	<i>tl</i>	654 743
jean	0.38	10	0.04	0	0.29	2	0.24	6
3-FullIns_3	<i>tl</i>	54	30.69	8 351	<i>tl</i>	190	2621.40	405 857
queen9_9	<i>tl</i>	3657	<i>tl</i>	692 134				
david	1927.06	10	0.03	0	21.26	5	0.08	0
mug88_1	0.44	1	54.96	39 195	0.62	1	<i>tl</i>	1 011 301
mug88_25	1.45	12	30.55	20 791	0.51	3	<i>tl</i>	1 225 319
1-FullIns_4	<i>tl</i>	8	33.70	3 425	<i>tl</i>	8	<i>tl</i>	162 882
myciel6	<i>tl</i>	4	72.64	7 904	<i>tl</i>	3	2000.39	183 177
queen8_12	<i>tl</i>	2768	<i>tl</i>	385 175				
mug100_1	2.09	14	113.58	99 468	2.71	33	<i>tl</i>	810 120
mug100_25	2.51	26	160.96	141 665	1.35	7	<i>tl</i>	1 081 903
queen10_10	<i>tl</i>	2507	<i>tl</i>	374 396	0.01	3	1.06	0
4-FullIns_3	<i>tl</i>	17	91.15	9 053	<i>tl</i>	17	<i>tl</i>	92 823
games120	<i>tl</i>	67	<i>tl</i>	763 536	<i>tl</i>	690	<i>tl</i>	262 064
queen11_11	<i>tl</i>	2120	<i>tl</i>	201 943	<i>tl</i>	8639	<i>tl</i>	39 063
r125.1	49.66	1	0.07	0	372.47	1	0.13	0
DSJC125.1	<i>tl</i>	14	<i>tl</i>	323 047	<i>tl</i>	14	<i>tl</i>	44 951
r125.5	<i>tl</i>	52	1992.94	178 069				
DSJC125.5	<i>tl</i>	14	<i>tl</i>	151 754	<i>tl</i>	141	<i>tl</i>	5 380
r125.1c	<i>tl</i>	12	1.21	32				
miles250	38.30	1	0.04	0	2.74	1	0.22	0
miles500	5.10	1	117.71	14 803	873.90	200	<i>tl</i>	242 578
miles750	<i>tl</i>	5	639.56	49 132	<i>tl</i>	155	<i>tl</i>	188 792
miles1000	<i>tl</i>	4	1012.08	200 247				
miles1500	<i>tl</i>	1	6.78	465				
anna	<i>tl</i>	8	0.15	0	<i>tl</i>	17	0.42	0
queen12_12	<i>tl</i>	1631	<i>tl</i>	96 305	<i>tl</i>	7733	<i>tl</i>	11 992
2-Insertions_4	<i>tl</i>	5	236.10	17 285	<i>tl</i>	4	<i>tl</i>	46 205
Solved	21/42		34/42		19/31		15/31	
karate	0.11	13	0.03	0	0.03	4	0.06	0
chesapeake	1.28	86	0.80	793	0.10	15	11.20	11 755
dolphins	0.72	1	0.29	30	0.07	4	8.91	2 887
lesmis	17.53	11	0.14	0	1.01	2	0.42	8
polbooks	2036.05	359	58.83	20 170	330.90	288	<i>tl</i>	631 201
adjnoun	<i>tl</i>	1	1.88	40	<i>tl</i>	10	139.90	7 030
football	<i>tl</i>	35	<i>tl</i>	615 634	<i>tl</i>	149	<i>tl</i>	189 697
Solved	5/7		6/7		5/7		5/7	

contrast tends to explore a much larger number of nodes (up to 365 825), and on average needs thousands of nodes.

- For $k = 10$, there are 31 2nd-DIMACS and 7 10th-DIMACS instances, 38 feasible instances in total. For 10 instances, no method could find the optimal solution within time limit; the Branch and Price could solve 24 out of 38 instances and is the fastest method in 18 cases; the Cplex + reduction could solve 20 out of 38 instances and is the fastest method in 10 cases; 8 instances are solved by Branch and Price while Cplex + reduction fails; 4 instances are solved by Cplex + reduction while Branch

Table 3Formulation performance comparison on the DIMACS instances ($k = 15$ and $k = 20$).

	$k = 15$				$k = 20$			
	Branch and Price		Cplex + reduction		Branch and Price		Cplex + reduction	
	Time	Nodes	Time	Nodes	Time	Nodes	Time	Nodes
2-Insertions_3	0.10	22	683.56	676 931				
myciel5	35.87	82	304.86	71 515	1.71	33	2434.94	1 105 526
2-FullIns_3	1.03	25	191.99	29 987	1.46	40	992.17	382 558
3-Insertions_3	0.51	14	<i>tl</i>	783 072	0.44	17	<i>tl</i>	766 107
1-Insertions_4	36.48	49	<i>tl</i>	143 540	12.33	50	<i>tl</i>	104 232
huck	0.14	4	14.76	6 619	0.07	2	4.98	1 745
4-Insertions_3	4.29	29	<i>tl</i>	369 502	3.13	38	<i>tl</i>	201 377
jean	0.63	8	0.64	0	0.33	7	5.20	2 085
3-FullIns_3	<i>tl</i>	79	<i>tl</i>	182 041	460.07	249	<i>tl</i>	105 272
david	0.27	2	14.75	4 111	1.73	18	<i>tl</i>	2 581 828
mug88_1	1.14	21	<i>tl</i>	540 602	0.85	13	<i>tl</i>	348 214
mug88_25	0.49	1	<i>tl</i>	459 082	1.10	33	<i>tl</i>	218 404
1-FullIns_4	<i>tl</i>	14	<i>tl</i>	72 206	<i>tl</i>	22	<i>tl</i>	25 797
myciel6	<i>tl</i>	3	<i>tl</i>	59 139	<i>tl</i>	6	<i>tl</i>	25 128
mug100_1	1.73	14	<i>tl</i>	413 118	1.51	12	<i>tl</i>	247 039
mug100_25	1.97	22	<i>tl</i>	376 089	1.64	18	<i>tl</i>	217 955
4-FullIns_3	<i>tl</i>	14	<i>tl</i>	69 442	<i>tl</i>	33	<i>tl</i>	16 830
games120	<i>tl</i>	1779	<i>tl</i>	139 497	<i>tl</i>	1	<i>tl</i>	81 290
ri25.1	0.96	1	275.62	35 565	2.32	1	<i>tl</i>	526 415
DSJC125.1	<i>tl</i>	15	<i>tl</i>	18 733	<i>tl</i>	22	<i>tl</i>	4 915
miles250	0.74	1	<i>tl</i>	614 993	1.85	8	<i>tl</i>	397 307
miles500	<i>tl</i>	1769	<i>tl</i>	156 294				
anna	57.52	7	0.78	17	84.68	7	2.06	30
2-Insertions_4	<i>tl</i>	8	<i>tl</i>	24 929	<i>tl</i>	8	<i>tl</i>	6 093
Solved	16/24		8/24		16/22		5/22	
karate	0.02	4	0.06	0	0.01	3	0.08	100
chesapeake	0.06	9	8.46	4 903				
dolphins	0.16	8	316.13	195 342	0.10	4	<i>tl</i>	1 688 935
lesmis	0.58	6	1.23	804	0.41	4	7.44	2 226
polbooks	25.94	44	<i>tl</i>	279 125	3.13	11	<i>tl</i>	267 568
adjnoun	<i>tl</i>	12	2337.24	157 113	<i>tl</i>	28	<i>tl</i>	90 669
football	<i>tl</i>	1544	<i>tl</i>	117 103	<i>tl</i>	9228	<i>tl</i>	57 614
Solved	5/7		5/7		4/6		2/6	

and Price fails. For the solved instances, the number of nodes explored by the Branch and Price is not larger than 288, Cplex + reduction explores up to 405 857 nodes, and on average needs much more nodes to solve the same graph instance for $k = 10$ than for $k = 5$.

- For $k = 15$, there are 24 2nd-DIMACS and 7 10th-DIMACS instances, 31 feasible instances in total. For 8 instances, no method could find the optimal solution within time limit; the Branch and Price could solve 21 out of 31 instances and is the fastest method in 20 cases; the Cplex + reduction could solve 13 out of 31 instances and is the fastest method in 2 cases; 9 instances are solved by Branch and Price while Cplex + reduction fails; 1 instance is solved by Cplex + reduction while Branch and Price fails. For the solved instances, the number of nodes explored by the Branch and Price is not larger than 82. Cplex + reduction needs to explore on average several thousands of nodes.
- For $k = 20$, there are 22 2nd-DIMACS and 6 10th-DIMACS instances, 28 feasible instances in total. For 8 instances, no method could find the optimal solution within time limit; the Branch and Price could solve 20 out of 28 instances and is the fastest method in 19 cases; the Cplex + reduction could solve 7 out of 28 instances and is the fastest method in 1 case; 12 instances are solved by Branch and Price while Cplex + reduction fails. For the solved instances, the number of nodes explored by the Branch and Price is not larger than 249. Cplex + reduction needs to explore on average several thousands of nodes and is able to solve only few instances. All instances solved by Branch and Price require less than 12.23 CPU seconds, except one that needs 460.07 s.

From these results we can conclude that **Cplex + reduction** has an average good performance for $k = 5$, and has increasing difficulties for larger values of k . A partial explanation can be found in the increase in the number of variables (n more variables for each incremental value of k). For $k = 5$, **Cplex + reduction** outperforms **Branch and Price**. For **Branch and Price**, an opposite behavior is experienced when increasing the value of k . In this case, the performance of the method is improved. For example, instance **polbooks** needs 2036.05 CPU seconds for $k = 5$, while 330.90, 25.94, and 3.13 CPU seconds are needed for $k = 10, 15$ and 20 , respectively. For $k = 10, 15$ and 20 , **Branch and Price** outperforms then **Cplex + reduction**.

Gaps. In Table 4 we report, for each value of k , the value of the optimal or best known solution (column opt^*), and the linear relaxation and optimality gaps for the 10th-DIMACS instances. The linear programming relaxation $lp\ gap$ is computed with respect to the optimal solution value opt as $100 \cdot \frac{lp_{val} - opt}{opt}$, where lp_{val} is the value of the linear programming relaxation of the corresponding formulation (or the best known solution). For instances for which the optimal solution value is not known, the $lp\ gap$ is not reported. A “–” is reported when the time limit is incurred before the linear relaxation is computed. The optimality gap $opt\ gap$ is computed as $100 \cdot \frac{UB_{val} - LB_{val}}{UB_{val}}$, where UB_{val} and LB_{val} are the values of the best upper bound and of the incumbent solution of the corresponding method when the time limit is reached (0.00 for solved instances). The same figures are omitted for the 2th-DIMACS instances, because they have a similar pattern. For the ILP_E , the $opt\ gap$ is computed using $LB_{val} = lp_{val}$ since the bound is never updated during the search tree which is explored in a depth-first fashion ($LB_{val} = UB_{val}$ for the instances solved to proven optimality).

From the table we observe that the formulation ILP_E is characterized by a much stronger linear programming relaxation. The value of its $lp\ gap$ is not affected by the value of k , and ranges between 0.0 and 6.43. This explains the fact that **Branch and Price** explores on average a much smaller number of nodes, and justifies the computational effort spent in column generation. On the other hand, the quality of the $lp\ gap$ of ILP_C deteriorates when k increases, and can be as large as 42.60. Clearly, computing this bound is associated with a smaller computational effort, and many nodes can be explored in short CPU time. For $k = 5$, the generic cuts embedded in the **cplex** MIP solver compensate the poor quality of the linear relaxation, while starting from $k = 10$ the increasing $lp\ gap$ cannot be effectively reduced and **Cplex + reduction** struggles in solving the associated instances.

Computational performance on intersection graph instances. For the Intersection Graph instances from [8] we concentrate on the values of $k = 15, 20$, where, according to the results of the previous section, the use of the Branch-and-Price algorithm showed to be more advisable. Results are summarized in Table 5 where we report, for **Branch and Price** and **Cplex + reduction**, the CPU time in seconds (tl for time limit) and the associated number of explored nodes. For each instance and for each value of k , we report in bold the fastest method. Missing lines correspond to infeasible instances.

- For $k = 15$ we considered 27 Intersection Graph instances in total. For 10 instances, no method can find the optimal solution within time limit; the **Branch and Price** can solve 11 out of 27 instances and is the fastest method in 9 cases; the **Cplex + reduction** can solve 11 out of 27 instances and is the fastest method in 10 cases; 7 instances are solved by **Branch and Price** while **Cplex + reduction** fails; 7 instances are solved by **Cplex + reduction** while **Branch and Price** fails. For the solved instances, the number of nodes explored by the **Branch and Price** is not larger than 6069. **Cplex + reduction** needs to explore on average several thousands of nodes.
- For $k = 20$, there are 16 feasible Intersection Graph instances. For 6 instances, no method can find the optimal solution within time limit; the **Branch and Price** can solve 6 out of 16 instances and is the fastest method in all the 6 cases; the **Cplex + reduction** can solve 5 out of 16 instances and is the fastest method in 4 cases; 5 instances are solved by **Branch and Price** while **Cplex + reduction** fails,

Table 4

LP relaxations and optimality gaps (DIMACS-10 instances).

	$k = 5$					$k = 10$				
	Branch and Price			Cplex + reduction		Branch and Price			Cplex + reduction	
	opt^*	$lp\ gap$	$opt\ gap$	$lp\ gap$	$opt\ gap$	opt^*	$lp\ gap$	$opt\ gap$	$lp\ gap$	$opt\ gap$
karate	32	1.42	0.00	5.25	0.00	30	1.76	0.00	10.99	0.00
chesapeake	32	6.43	0.00	17.54	0.00	27	4.26	0.00	30.13	0.00
dolphins	60	0.00	0.00	3.22	0.00	55	0.00	0.00	11.29	0.00
lesmis	76	0.56	0.00	1.26	0.00	75	0.88	0.00	2.57	0.00
polbooks	97	3.21	0.00	7.61	0.00	90	4.50	0.00	14.28	4.63
adjnoun	110	—	52.68	1.77	0.00	106	0.00	50.00	5.34	0.00
football	94		10.88		6.40	71		22.09		30.00

	$k = 15$					$k = 20$				
	Branch and Price			Cplex + reduction		Branch and Price			Cplex + reduction	
	opt^*	$lp\ gap$	$opt\ gap$	$lp\ gap$	$opt\ gap$	opt^*	$lp\ gap$	$opt\ gap$	$lp\ gap$	$opt\ gap$
karate	28	1.75	0.00	16.49	0.00	23	1.43	0.00	28.31	0.00
chesapeake	22	0.90	0.00	42.60	0.00					
dolphins	49	1.41	0.00	20.97	0.00	43	2.16	0.00	30.64	6.32
lesmis	74	0.80	0.00	3.85	0.00	72	0.69	0.00	6.41	0.00
polbooks	86	2.57	0.00	18.08	6.88	80	2.79	0.00	23.79	11.26
adjnoun	101	0.00	47.52	9.81	0.00	96		1.04		3.87
football	54		37.04		42.75	44		23.76		39.25

Table 5Formulation performance comparison on the Intersection Graph Instances ($k = 15$ and $k = 20$).

	$k = 15$				$k = 20$			
	Branch and Price		Cplex + reduction		Branch and Price		Cplex + reduction	
	Time	Nodes	Time	Nodes	Time	Nodes	Time	Nodes
bcspwr02	1.38	1	0.78	381				
impcol	23.85	181	22.49	29 240	25.89	689	380.80	717 423
dwt	22.19	253	154.07	63 767				
can62	22.09	7	<i>tl</i>	1 118 811				
dwt72	121.35	6 069	<i>tl</i>	903 196	466.80	50 711	<i>tl</i>	685 004
ash219	114.16	779	<i>tl</i>	358 161	258.45	1 974	<i>tl</i>	174 120
dwt87	<i>tl</i>	4 587	202.22	43 774				
Solved	6/7		4/7		3/3		1/3	
ash331	<i>tl</i>	2 026	<i>tl</i>	237 020	<i>tl</i>	5 112	<i>tl</i>	143 100
gre	<i>tl</i>	261	<i>tl</i>	137 169	<i>tl</i>	574	<i>tl</i>	69 700
bcspwr03	1853.68	2 292	<i>tl</i>	260 470	<i>tl</i>	3 175	<i>tl</i>	187 362
L125.will199	44.21	41	<i>tl</i>	186 330	31.37	23	<i>tl</i>	95 182
L125.west0167	37.49	5	<i>tl</i>	324 965	30.43	2	<i>tl</i>	217 315
L125.ash608	<i>tl</i>	128	<i>tl</i>	173 467	<i>tl</i>	507	<i>tl</i>	99 585
L125.can	<i>tl</i>	1 495	<i>tl</i>	66 494	<i>tl</i>	15 705	<i>tl</i>	24 735
L125.gre	<i>tl</i>	492	<i>tl</i>	28 888				
L125.lop163	<i>tl</i>	346	<i>tl</i>	25 323				
L125.dwt	<i>tl</i>	11 848	<i>tl</i>	55 178				
L125.can	<i>tl</i>	2 430	<i>tl</i>	45 321				
west0132	41.57	12	<i>tl</i>	335 734	110.40	269	<i>tl</i>	225 030
rw136	<i>tl</i>	102	<i>tl</i>	33 405	<i>tl</i>	274	<i>tl</i>	14 098
Solved	4/13		0/13		3/9		0/9	
L80.cavity01	<i>tl</i>	1	4.96	2 107	<i>tl</i>	1	4.41	1 189
L80.wm1	85.64	1	0.40	39				
L100.cavity01	<i>tl</i>	1	5.50	2 093	<i>tl</i>	1	5.74	641
L100.wm1	<i>tl</i>	1	1.49	215				
L100.wm3	<i>tl</i>	12	4.21	1 528				
L120.cavity01	<i>tl</i>	1	5.44	542	<i>tl</i>	1	8.04	1 163
L120.wm2	<i>tl</i>	1	0.71	0	<i>tl</i>	1	46.99	17 695
Solved	1/7		7/7		0/4		4/4	

Table 6

LP relaxations and optimality gaps (Intersection graph instances).

	$k = 15$					$k = 20$				
	Branch and Price			Cplex + reduction		Branch and Price			Cplex + reduction	
	opt^*	$lp\ gap$	$opt\ gap$	$lp\ gap$	$opt\ gap$	opt^*	$lp\ gap$	$opt\ gap$	$lp\ gap$	$opt\ gap$
bcspwr02	25	3.43	0.00	47.74	0.00	21	27.12	0.00	63.07	0.00
impcol	36	7.57	0.00	38.26	0.00					
dwt	18	24.63	0.00	69.22	0.00					
can62	35	2.23	0.00	43.47	13.27					
dwt72	46	6.06	0.00	36.11	27.62	36	10.94	0.00	49.99	34.77
ash219	59	5.57	0.00	30.59	24.57	51	6.16	0.00	40.00	34.72
dwt87	33	22.81	34.50	61.17	0.00					
ash331	69	9.49	18.67	33.65	28.14	55	16.58	21.13	47.12	38.29
gre	77	10.24	12.57	33.01	26.30	63	16.39	20.37	45.19	35.54
bcspwr03	83	3.11	0.00	29.66	20.58	69	6.58	18.76	41.51	29.06
L125.will199	105	1.50	0.00	16.00	10.79	98	1.61	0.00	21.60	14.91
L125.west0167	108	0.31	0.00	13.60	9.70	101	0.66	0.00	19.20	13.84
L125.ash608	95	2.74	8.88	24.00	20.21	81	7.61	14.46	35.20	31.50
L125.can	42	37.96	46.83	66.40	57.96	23	51.31	57.66	81.60	67.38
L125.gre	38	48.77	67.65	69.59	66.75					
L125.lop163	28	61.65	76.72	77.60	75.70					
L125.dwt	45	32.35	75.95	64.00	53.41					
L125.can	23	61.86	75.12	81.58	76.80					
west0132	111	1.07	0.00	15.90	10.86	103	1.15	0.00	21.96	16.19
rw136	101	4.91	7.73	25.73	23.10	86	8.99	54.49	36.76	33.23
L80.cavity01	60	–	–	23.96	0.00	49	–	–	37.64	0.00
L80.wm1	31	3.07	0.00	53.96	0.00					
L100.cavity01	79	–	–	20.19	0.00	68	–	–	31.05	0.00
L100.wm1	52	–	–	44.55	0.00					
L100.wm3	47	4.90	4.90	47.99	0.00					
L120.cavity01	97	–	–	17.95	0.00	88	–	–	25.39	0.00
L120.wm2	107	–	–	9.76	0.00	79	–	–	32.02	0.00

4 instances are solved by **Cplex + reduction** while **Branch and Price** fails. For the solved instances, the number of nodes explored by the **Branch and Price** can be as large as 50 711, while **Cplex + reduction** can need one order of magnitude additional nodes for solved instances.

From these results we can conclude that for the Intersection Graph instances **Cplex + reduction** and **Branch and Price** have similar performance. However, the two solution methods are highly complementary: there are 23 instances that can be solved to optimality by one method only.

Gaps. In Table 6 we report, for each value of k , the value of the optimal or best known solution (column opt^*), and the linear relaxation and optimality gaps for the Intersection Graph instances. From the table we observe that, also for the Intersection Graph instances, the formulation ILP_E is characterized by a much stronger linear programming relaxation, although the gap appears to be increasing with the value of k . The strong linear programming relaxation explains the fact that **Branch and Price** explores on average a much smaller number of nodes, and justifies the computational effort spent in column generation. Again, the quality of the $lp\ gap$ of ILP_C deteriorates when k increases, but instances with a $lp\ gap$ as large as 63.07% can still be solved to optimality, thanks to the capability of the **cplex** MIP solver to process a large number of nodes in short computing time.

6. Conclusions

In this paper we considered the minimum vertex k -cut problem, a variant of graph partitioning which consists in finding a vertex k -cut of minimum cardinality. We studied two alternative ILP formulations and analyzed their properties in terms of linear programming relaxation. The first formulation is a natural compact formulation while the second one is an exponential-size formulation which requires Column

Generation techniques to be effectively solved. We proposed a Branch-and-Price algorithm and we showed how to solve the Linear Programming relaxation of the exponential-size formulation in polynomial time via a series of Min-Cut Max-Flow problems. We computationally compared the performances of the two formulations on benchmark instances from the literature. The outcome of these experiments is that the Branch-and-Price algorithm outperforms the direct use of a general-purpose ILP solver on the compact formulation for large values of k (high number of disconnected subsets of the partition). For small values of k instead, directly tackling the compact formulation remains the best option.

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Appendix

A.1. Pricing as submodular function minimization

The $\text{PP}\cup\emptyset$ (i.e., the relaxation of the PP where also the empty set is admitted as solution) can also be tackled as the minimization of a submodular function and hence is polynomial-time solvable [29]. A list of submodular functions is reported in a list in [29] (section 44.1a).

First, given $S \subseteq V$, let us define the two following clique families:

$$I(S) := \{C \in \mathcal{C} : C \cap S \neq \emptyset\} \quad \text{and} \quad C(S) := \{C \in \mathcal{C} : C \subseteq S\}.$$

Second, we use the short-hand notation:

$$\nu^*(S) := \sum_{v \in S} \nu_v^* \quad \pi^*(\mathcal{C}') := \sum_{C \in \mathcal{C}'} \pi_C^*.$$

The $\text{PP}\cup\emptyset$ can be formulated as

$$\max_{S \subseteq V} \nu^*(S) - \pi^*(I(S)).$$

Observe that $C \in I(S)$ if and only if $C \notin C(\bar{S})$. Hence a set S maximizes $\nu^*(S) - \pi^*(I(S))$ if and only if its complementary set minimizes the set function

$$f(S) := \nu^*(S) - \pi^*(C(S)). \tag{22}$$

Proposition 6 implies that the set function $f(\cdot)$ is both submodular and supermodular.

Proposition 6. $f(S) + f(T) = f(S \cap T) + f(S \cup T)$, for every $S, T \subseteq V$.

Proof. Obviously, $\nu^*(S) + \nu^*(T) = \nu^*(S \cap T) + \nu^*(S \cup T)$. Clearly, $\pi^*(C(S)) + \pi^*(C(T)) = \pi^*(C(S \cap T)) + \pi^*(C(S \cup T))$. $\square$

A.2. Pricing as a min-cut on a smaller network

In case $\mathcal{C}$ is exactly the set of edges of the graph G (which, for instance, the only possible form of $\mathcal{C}$ for a triangle free graph), a solution method based on solving a min-cut problem on a smaller network can be

exploited. First observe that, since the cliques in $\mathcal{C}$ are in fact the edges of G , then $I(S) \setminus C(S) = \delta(S)$ for all $S \subseteq V$. Furthermore, (22) can be rewritten in standard notation as $f(S) = \nu^*(S) - \pi^*(E(S))$, and since

$$2\pi^*(E(S)) + \pi^*(\delta(S)) = \sum_{v \in S} \pi^*(\delta(v)),$$

the $\text{PPU}\emptyset$ is equivalent to minimizing $2\nu^*(S) + \pi^*(\delta(S)) - \sum_{v \in S} \pi^*(\delta(v))$. Observe that the equation below holds where the third term in the last expression is a constant:

$$2\nu^*(S) + \pi^*(\delta(S)) - \sum_{v \in S} \pi^*(\delta(v)) = 2\nu^*(S) + \pi^*(\delta(S)) - \sum_{v \in V} \pi^*(\delta(v)) + \sum_{v \in S} \pi^*(\delta(v))$$

Hence, actually, the $\text{PPU}\emptyset$ amounts to

$$\min 2\nu^*(S) + \pi^*(\delta(S)) + \sum_{v \in S} \pi^*(\delta(v)).$$

This problem can be solved as a min-cut problem on network with source node s , sink node t , one node for each $v \in V$ and the arc set defined below:

- For each $v \in V$, there is an arc (s, v) with capacity $2\nu_v^*$;
- For each edge $uv \in E$, there are an arc uv and an arc vu with capacity π_{uv}^* ;
- For each node $v \in V$, there is an arc (v, t) with capacity $\pi^*(\delta(v))$.

In this case, the $\text{PPU}\emptyset$ can be solved in polynomial-time as a min-cut problem (equivalent to max-flow) on the above described network, having $n + 2$ nodes.

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